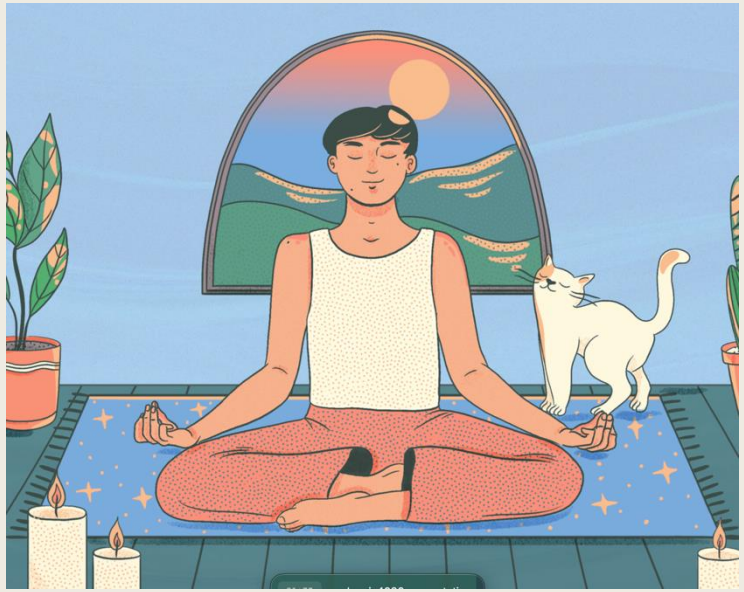
# EEG-Based Mind-Wandering Detection in Meditators

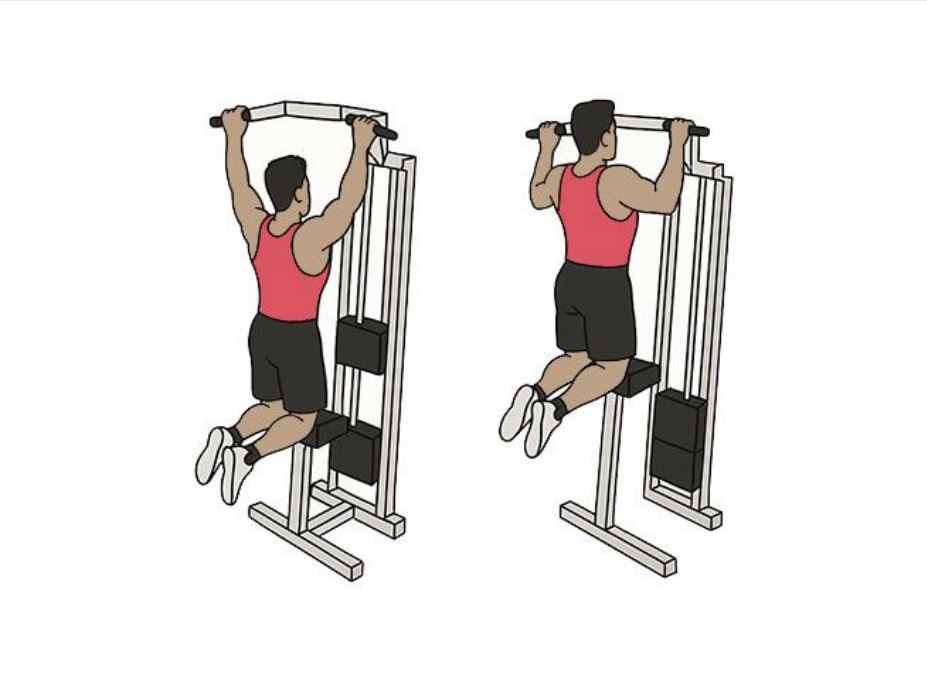
By Max Boyington

# Motivation Pt.1

- Mental health challenges are increasing worldwide
- Regulating attention is getting harder in an always-online, distraction-heavy world
- Meditation is a promising tool for improving focus and emotional regulation

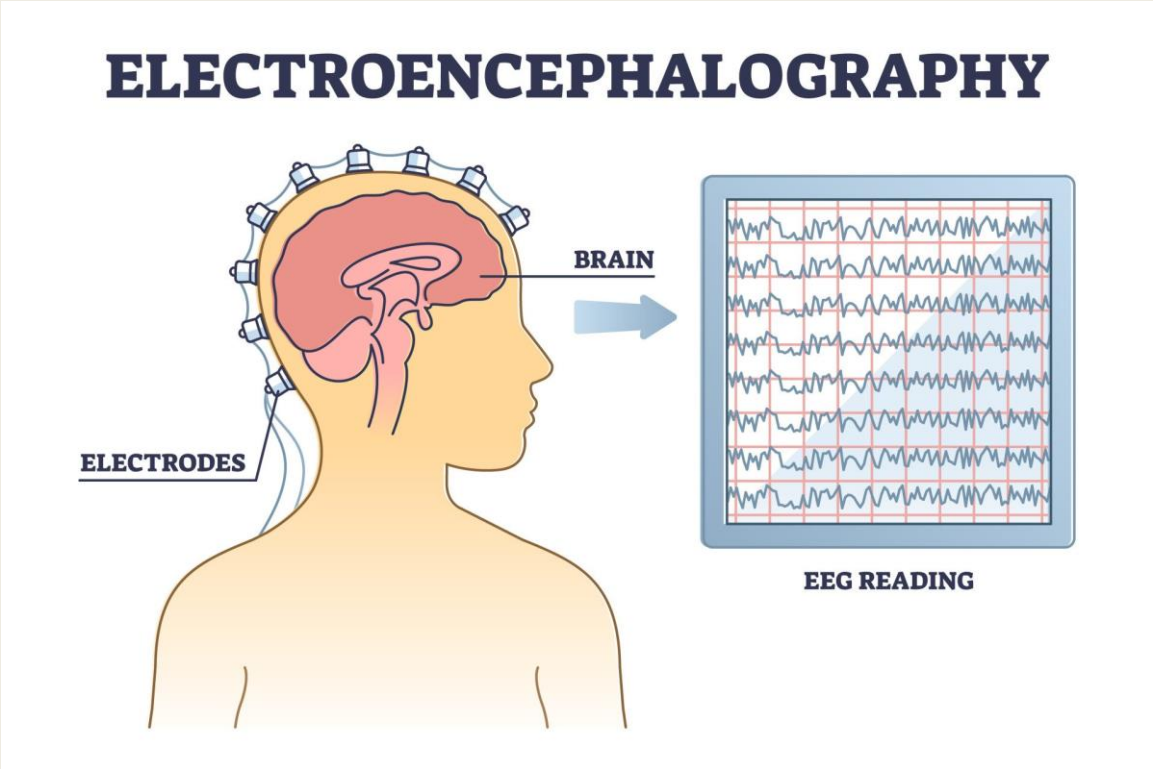


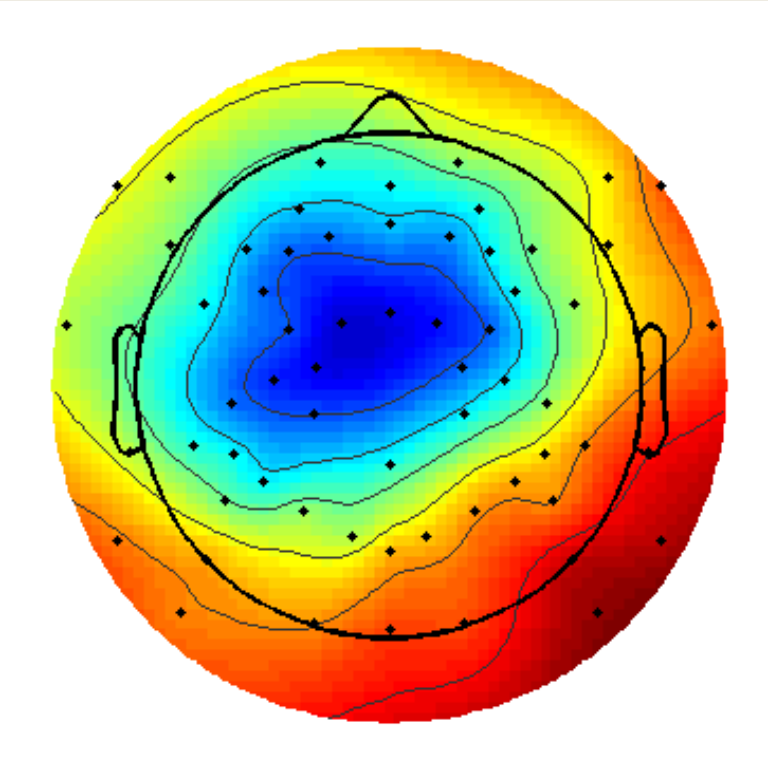
# Motivation Pt.2



- Meditation is hard to start and even harder to sustain, especially for those who need it most
- Assisted meditation could make it more approachable
- Some meditation products make strong claims without strong evidence that they can help this problem

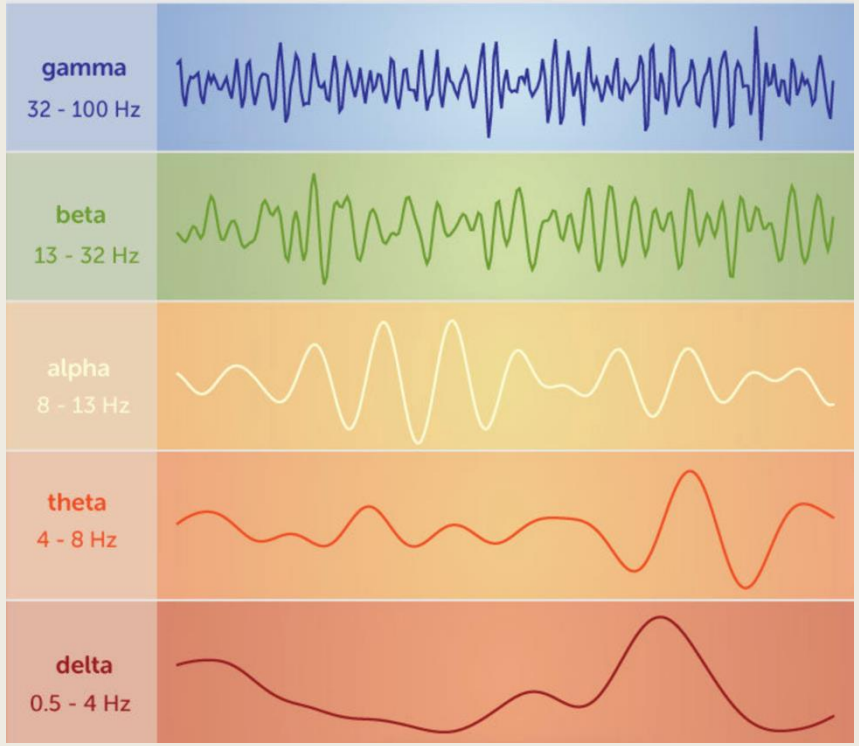
# What is EEG?



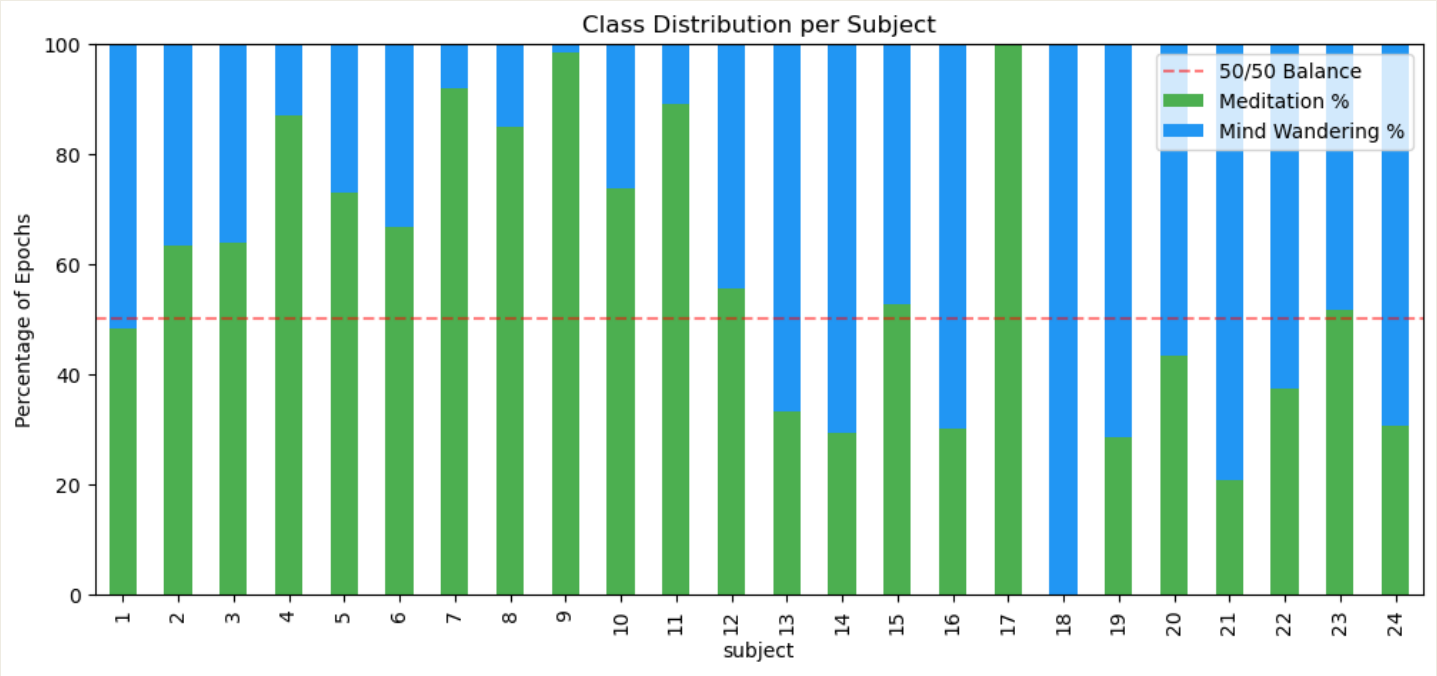
# The Research Questions

1. Evaluate how well an SVM & XGBoost generalize to **unseen subjects** versus **partially seen (calibrated) subjects** *(an underexplored setting in mind-wandering classification literature)*
2. Compare the **predictive value of different EEG features** *(to replicate findings in other papers)*[2,3]

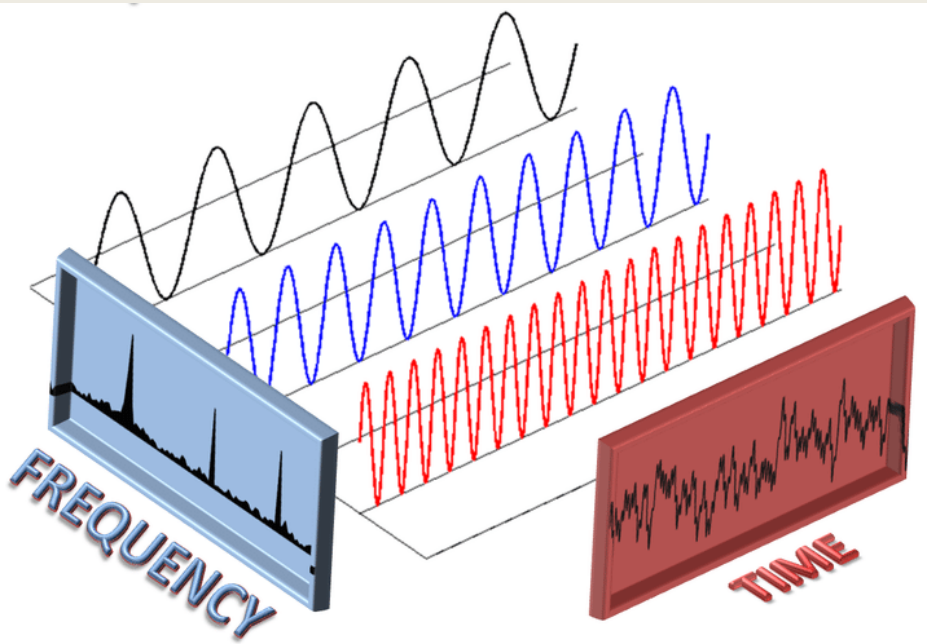


# The Dataset

- Used from public dataset from Tracy Brandmeyer's paper on mindwandering[1]
- **21 subjects** (after removing 3 subjects with extreme class imbalance)
- **64 electrode** EEG system
- 45-90 min of meditation (focusing on breath)
- Probed randomly every 30-90s to rate depth of meditation, mind-wandering, and drowsiness
- Used **10s epoch** prior to probe, as they did in the paper[1]
- Meditating class: **466 samples**
- Mind-wandering class: **296 samples**



# Feature Extraction




NOTE: This step happens times **64 electrodes**

1. The Input: Raw 10-sec EEG Time-Series Epochs
2. Transforming to a Power Spectral Density: **Fourier transformation** across 2-40Hz
3. Feature Extraction:
    1. Mean band power across Delta, Theta, Alpha, Beta, Gamma:
        - **Absolute power** (regular avg)
        - **Relative power** (relative to other avg band powers)
        - **Aperiodic-flattened power** (subtract 1/f slope from PSD)
    2. Aperiodic Components: **Exponent** and **offset**

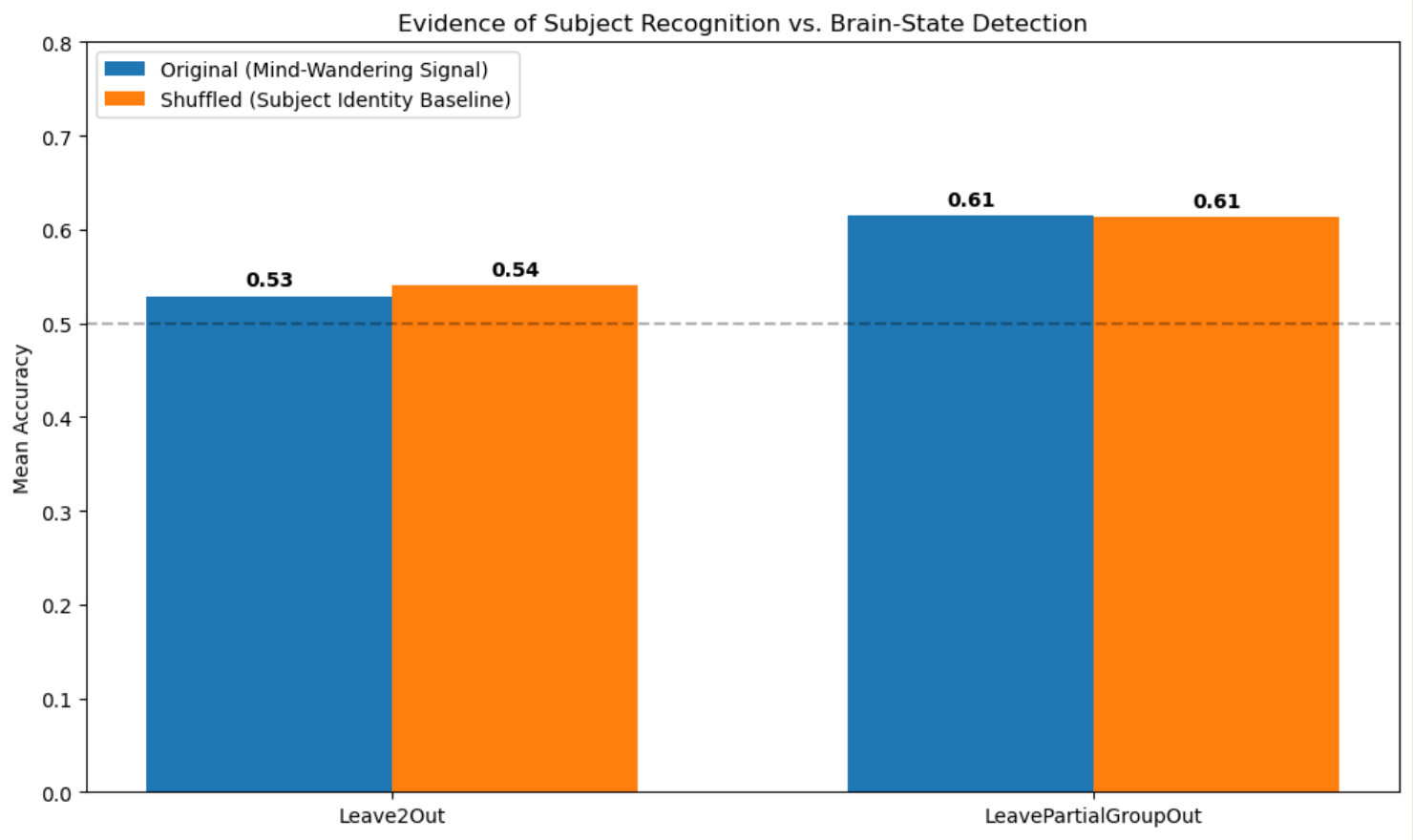
- Totals to **17 different feature types** (5 bands x 3 types of avg power, 2 aperiodic components) across **64 electrodes** (aka 1088 total feats)

# The Model

- **Support Vector Machine (SVM)** was selected due to the small dataset and that it performed strongest in other papers[2, 3]
  - RBF kernel, balanced class weight (to fix class imbalance), StandardScaler, PCA (n=10)
- **XGBoost** was selected for its robustness with tabular/non-linear data
  - 100 estimators, max depth of 2, StandardScaler, PCA (n=10)
- Evaluated using **F1 Macro** scoring to account for class imbalance
- To ensure models didn’t memorize individual subjects, they were also run on data where labels were shuffled across subjects

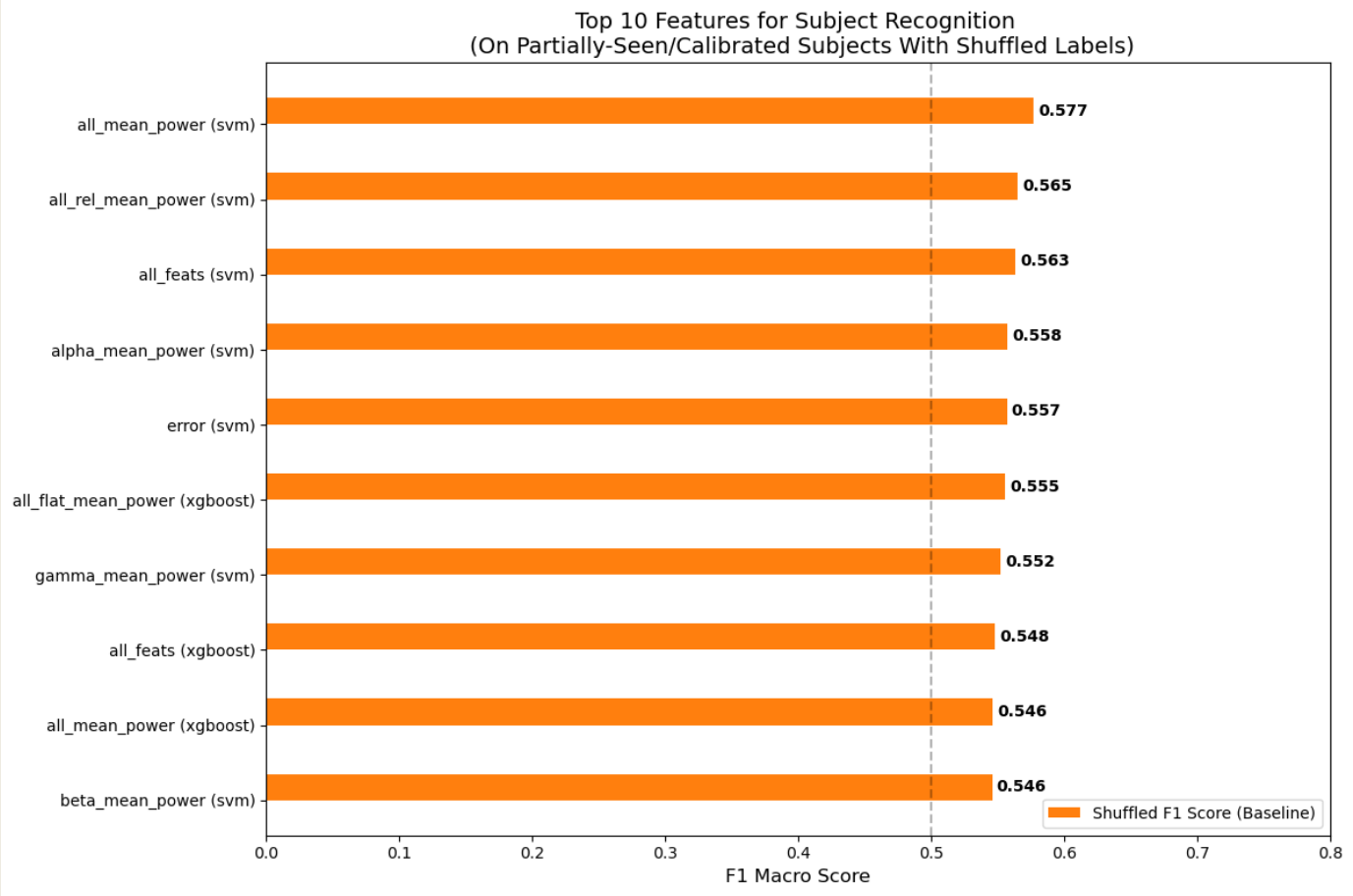
# The Results pt.1: Shuffled Labels Made No Difference in Performance

- Shuffled vs unshuffled labels across subjs were the same
- This means models were memorizing subjects based on their eeg fingerprints
- Model was an effective subject classifier but not as a brain-state classifier



# Results pt.2

- All mean power freq bands performed the best at detecting individual subj patterns
- Overall it seems the more feats, the more the model can detect subjs



# Limitations

- Dataset: Small dataset (n=21, 762 datapoints)
  - Inter-subject variability is likely very large with this type of data as it varies a lot between people (mind-wandering vs meditation)
- Preprocessing: Minimal preprocessing pipeline could not remove artifacts like eyeblinks and muscle tension
- Features: May be too simple; PSD features flatten 10s of time series data into only one number
- Models: May also be too simple; SVMs likely can’t recognize the nuances of EEG data

# Future Directions

- Dataset: Transitioning to a larger dataset of n=125 (4,500 datapoints)
- Preprocessing: Use more robust preprocessing pipeline (new dataset has scripts included)
- Features: Move from time-collapsed PSD data to time-series
- Models: Deep learning (CNNs or LSTMs, or graph-based models)

# Bibliography


1. Brandmeyer, Tracy, and Arnaud Delorme. “Reduced Mind Wandering in Experienced Meditators and Associated EEG Correlates.” *Experimental Brain Research* 236, no. 9 (2018): 2519–28. https://doi.org/10.1007/s00221-016-4811-5.
2. Jin, Christina Yi, Jelmer P. Borst, and Marieke K. Van Vugt. “Predicting Task-General Mind-Wandering with EEG.” *Cognitive, Affective, & Behavioral Neuroscience* 19, no. 4 (2019): 1059–73. https://doi.org/10.3758/s13415-019-00707-1.
3. Tang, Shaohua, Yutong Liang, and Zheng Li. “Mind Wandering State Detection during Video-Based Learning via EEG.” *Frontiers in Human Neuroscience* 17 (May 2023): 1182319. https://doi.org/10.3389/fnhum.2023.1182319.